

Institute of Actuaries of India

Subject SA5 – Finance

March 2017 Examination

INDICATIVE SOLUTION

INTRODUCTION:

The indicative solutions provided are very detailed in nature for the benefit of students and the examiners are not expecting the students to provide such in-depth details under examination conditions. However the students are expected to cover all the fundamental principles to demonstrate their understanding of the subject.

Solution 1:-

i) The factors to be considered will include

- Cost of the raw material for growing “Bharath Herbicia”.
- Average produce expected year-over-year and the anticipated sales revenue.
- The additional risks of growing this plant vis-à-vis the other crops which can be grown
- Elasticity of demand for “Bharath Herbicia” vis-à-vis other crops
- Incremental costs of processing the plant for its juice and the cost of preparation
- Requirements related to infrastructure and logistics.
- Supply chain – forward and backward linkages.

(4)

ii) The key stages in the RAMP (Risk Analysis and Management of Projects) process are as follows:

- Process Launch: The farmer needs to set up a team (which can include himself and other key stakeholders) to examine the project which is growing the medicinal plant called “Bharat Herbicia”. This team will be responsible for managing the RAMP process from the start to end.

This team will ensure that the following areas are identified and documented:

- Key objectives including financial objectives like the required rate of return from the project
- Scope of the project
- Magnitude and timing of investment
- Use of human resources
- Assessment of the strategic value of the project

- Risk Review

The risk review process is carried out at the outset and then regularly over the life of the project. The farmer needs to determine the time horizon over which the risk review needs to take place.

This stage involves

- Identifying all significant risks arising from growing “Bharat Herbicia”:
- Appraising these risks and grouping similar risks – for example risks with a common cause
- Evaluating/assessing these risks including the interactions between these risks. A financial model is developed to assess the impact of these risks on the projected future cash flows associated with the project.

This stage also involves identifying both catastrophic outcomes as well as favourable outcomes together with the likelihood of these outcomes.

This stage will culminate in designing a risk mitigation strategy.

Risk mitigation can happen through

- Risk reduction
- Risk transfer e.g., crop insurance

- Risk avoidance e.g., growing a medicinal plant, which is different from “Bharat Hebraic” because the other type of medicinal plant is more widely used in the preparation of medicines.
- Risk Management
This phase of RAMP involves (a) implementation of risk mitigation strategy; and (b) management of any remaining risks that have not been fully mitigated.
- Typically, unmitigated risks are managed through
 - Containment plans to avoid such risks
 - Contingency plans in the event that the risk occurs
 - Contingency budget to deal with the effects of the unmitigated risks
- Process Close down
At the end of the project, the RAMP process is reviewed. The aim of this phase is to learn from the experience of the project in order to refine the RAMP process for future use.

(9)

iii) The key risks will include:

- Demand for the Ayurveda Medicines which use this plant as an ingredient
- Quality of the juice produced & the technology used for extraction & preparation
- Relationship with buyers including contracts for guaranteed purchase
- Liquidity Risk – ability to generate cash as and when needed
- Research & Development Efforts related to developing new uses for the plant
- Shifts in consumer preferences for non – Ayurveda medicines

(6)

iv) The matrix can be used for

- Projecting revenues
- Assessing the risk of not being able to meet the demand of buyers for different grades for the plant
- Drawing up a risk management strategy for maximising the number of Grade A & Grade B plants & minimising the numbers of plant deaths.
- Detecting adverse changes in transition probabilities at an early stage & putting in place appropriate risk mitigation strategies

(5)

v) Invention refers to the discovery of a new technique while innovation refers to the implementation of the new technique.

- Growing “Bharath Herbicia” may qualify as an innovation if the organic farmer is implementing a new technique for growing the medicinal plant
- Appraising inventions and innovations can be challenging because such projects will have a greater degree of uncertainty than most capital projects.
- One of the challenges is to estimate the true opportunity cost of capital to be used for discounting the expected cash flows from the project. This is because the new venture may have a different level of systematic risk (different beta) as compared to the systematic risk of the farmer’s normal agricultural operations.

- It is also important to note that inventions and innovations can have economic rents.

Examples of such economic rents are:

- Supernormal profits being made for a period of time due to the high demand for the product and the farmer is the only producer
- The protection offered by patents and proprietary technology or production cost advantages compared to the competitions
- Possible contractual advantages – for example the farmer is the exclusive supplier to some of the major firms engaged in manufacturing Ayurveda medicines.

While evaluating economic rents, the farmer needs to carefully understand both the nature of the economic rent and the time period for which such rents can be enjoyed.

- The farmer should consider the impact (possibly detrimental) of this new venture on his other lines of business.
- The farmer should also consider and carefully value any real options associated with this project – for example, are there some “follow-on” investment opportunities associated with this project? If so, then the true value of the project will be equal to the discounted cash flow value of the project plus the value of the option to make any follow-on investments.

(7)

vi) Liquidity risk refers to the risk of money markets not being able to supply funding to business when required or more broadly to the management of short-term cash flow requirements. Alternatively it can refer to the insufficient capacity in the market to handle asset transactions at a time when the deal is required (without a material impact on the price).

(1)

vii) The sources of liquidity risk will be all those factors which can adversely impact the working capital cycle (net operating cycle) of this business.

- a)** Slow Moving Inventory: The inventory turnover (defined as sales divided by average inventory) may turn-out to be lower than expected – probably due to a fall in the demand for the Ayurveda medicines which use Bharat Herbicia as an ingredient. In that case more money is locked up in inventory creating liquidity issues
- b)** Longer Collection Period: The customers may delay payment beyond the stipulated credit period and this can result in a situation of insufficient liquidity
- c)** Lack of Trade Credit: The suppliers of raw materials & utilities may insist on cash payment on delivery or may even demand advance payment which can create a liquidity strain.
- d)** Restricted Bank Borrowings: The banks may be reluctant to extend liberal credit facilities for farmers growing medicinal plants which can lead to a liquidity crunch.

The farmer can manage the liquidity risk by

- a) Persuading the banks to provide a liberal line of credit by submitting a robust business plan
- b) Approaching micro-finance institutions which will have more appetite for financing such ventures.
- c) Unlocking the funds blocked in receivables through arrangements like advance factoring where the factor will provide a short-term loan against the receivables which have been factored.
- d) Entering into partnership models with the customers – manufacturers of Ayurveda medicines – so that the customers have a stake in ensuring that the farmer's venture is successful and is not plagued by liquidity crisis.

(5)

viii) The valuation for the business needs to take into account the following factors:

- Projected earnings (EBITDA – Earnings before Interest, Taxes & Depreciation) for the next 5-10 years or over the estimated useful life of the project
- Projected post-tax cash flows in terms of the initial outflow, operating cash flow & terminal cash flow.
- Identification, Assessment & Management of the key risks associated with the project – particularly the residual risks that cannot be mitigated.
- Comparison with the peer group engaged in growing similar medicinal plants – their business models and earning & profitability track record.
- Access to technology know-how in growing and processing the medicinal plants.
- Government policy for the next few years –for example, marginal tax rates, subsidy, tax breaks for R&D spend, etc.
- Cost of the land and value of the land for the intended purpose
- Valuation Metrics – P/E (Price - Earnings) Multiple Approach, P/B (Price – Book Value) Multiple Approach or DCF (Discounted Cash flow Approach) or a combination of these approaches.
- Validation of key financial assumptions like the basis for determining the discount rate & the future earnings growth rate and validation of key technical assumptions like the transition probabilities. (Probabilities of changing states).

The exit options available to the venture capitalist will include:

- Sell to the existing shareholders at a mutually agreed price subject to regulatory approvals
- Look out for another external investor
- Subject to the consent of other shareholders, consider listing the company on the stock exchanges
- Entering into a share buyback arrangement with the company
- Milestone based exit based on agreement with other shareholders.

(12)

ix) The factors considered by the Ministry will include

- Beneficial effects of the medicines using “Bharat Herbicia” as an ingredient
- Harmful side effects of the medicines using “Bharat Herbicia” as an ingredient
- Cost of these medicines vis-à-vis comparable medicines from alternative systems like allopathic drugs
- Whether these medicines have been certified as “safe for use” by relevant authorities – for example; the Central Drugs Standard Control Organisation (CDSCO) in India or the Food & Drug Administration (FDA) of the United States
- Export Potential for the medicines which will use “Bharat Herbicia” as an ingredient
- Harmful Externalities of growing “Bharat Herbicia” like the adverse impact it may have on the ecological balance of the region where it is grown
- Beneficial externalities like the employment potential it can generate for the local farming community
- Opportunity cost of the resources (like land) used for growing “Bharat Herbicia”.

(6)

[55 Marks]

Solution 2:-

i) The factors will include:

- Net worth and ability to comply with the capital adequacy requirements on an ongoing basis.
- Management Expertise particularly in terms of the technical expertise related to agriculture & allied operations and rural credit expertise.
- Risk Management framework for identifying, assessing and managing risks
- Ability to attract & retain appropriate talent
- Credit appraisal systems
- Quality of Corporate Governance
- Commitment to comply with regulations.
- Business Plan Projections including the sensitivity & scenario testing which have been carried out in respect of the plan projections.

(7)

ii) The key factors to be considered are

- Land Ownership
- Ability to Provide Collateral
- Historical Yields on the Land
- Credit repayment track – record.
- Savings History & Savings Propensity of the local community
- Farming Methods – the extent to which these methods have kept pace with the developments in agricultural technology

- Extent to which the farmers are willing to form credit groups where if one farmer defaults on the forming obligations, the other farmers will have to pitch in

(7)

- iii) A micro-finance institution can face significant concentration of credit risk because the loans are extended to the local farming community & small business units.

Even if the loans are small and extended to a variety of customers (borrowers), the credit risks will all be correlated.

Many of the loans are likely to default at once if the local economy experiences a downturn –for example due to monsoon failure

Use of securitization:

The most straight - forward instrument which can be used for reducing the concentration of credit risk will be an asset - backed security (ABS).

Under the ABS structure, the micro-finance institution will ring – fence a portfolio (pool) of loans and assign the future cash flows from these loans to a SPV (special purpose vehicle).

The SPV will then sell bonds to investors who receive interest & capital payments based on the cash flows arising from the ring – fenced pool of loans.

In the event of default by the borrowers, the micro-finance institution will no longer suffer loss because the credit (default) risk is transferred to the investors.

The micro-finance institution can utilize the cash paid by the investors (through the SPV) to invest in assets that are not correlated to its current loan portfolio & thereby reduce the overall credit risk exposure.

A more complex variant of the simple ABS is the CDO (collateralized debt obligation) which has multiple tranches of debt securities. These tranches differ in terms of the priority of capital repayment in the event of default in the underlying loan portfolio.

(6)

- iv) Credit Rating (also referred to as debt rating) essentially reflects the probability of timely payment of interest and principal by a borrower.

The higher the credit rating, the greater the likelihood that the borrower will fulfil his obligations to pay the interest and the principal.

The limitations of credit rating are:

A credit rating is not a recommendation for purchasing, selling or holding a debt instrument. The important elements relevant for investment decision-making in a debt instrument are (i) yield to maturity; (ii) risk tolerance of the investor; and (iii) credit risk of

the instrument. Since credit rating focuses only on the credit risk of the instrument, it cannot be the sole basis for investment decision-making.

- A credit rating is not a general /overall evaluation of the issuer. If a debt issue of firm X is rated higher than the debt issue of firm Y, it does not mean that firm X is better than firm Y.
- A credit rating does not create any sort of fiduciary relationship between the rating agency and the users of such ratings because there is no legal basis for such a relationship.
- A rating agency bases its rating on the information shared by the issuer and is not expected to audit or attest to the veracity of such information.
- A debt rating is not a one-time evaluation of credit risk which can be regarded as valid for the remaining term to maturity of the debt instrument. Hence as & when the business & financial conditions change, the ratings can change. However it needs to be noted that there can often be a significant time lag between the time the conditions change and the time when the ratings are revised.

(5)

v) The factors to be considered will include:

- The purpose for which the credit rating is sought
- Historical performance over the preceding 3 to 5 years
- If there have been significant changes in the business & financial conditions over the recent past, then how the company has managed to navigate these changes
- Quality of the credit Appraisal Process, Trends in NPAs (Non-Performing Assets) and profile of the customers (borrowers).
- Extent to which there is a concentration of credit risk
- Business Plan Projections and the bank's ability to meet the plan targets
- Regulatory changes and the impact of these changes on the bank's performance
- Debt Servicing Ability based on a detailed analysis of the bank's interest coverage ratios, liquidity ratios, etc.
- Extent of exposure to interest rate risk and bank's ALM (Asset Liability Management) policies to manage this risk
- Investment Policies of the Bank and Investment Performance
- Critical Observations of the Central Bank as reflected in the past inspections reports
- Quality of Management in terms of its commitment, experience and capacity to manage the key risks to which the bank is exposed
- Risk culture of the bank
- Quality & Composition of the Board; Effectiveness of the Board Sub-Committees like the Audit Committee & the Risk Committee; role & contribution of the Independent Directors
- Corporate Governance Framework
- Quality of Financial Disclosures

- Impact of the likely changes in the political, regulatory & economic environment on the performance & financial stability of the bank
- Exposure to operational risks and the bank's approach for managing these risks.

(8)

vi) Sources of Interest Rate Risk:

- Interest rate risk exists due to the fact that the assets and liabilities are of different terms
- For a bank like BRB, the typical situation will be one where the duration of the asset (loan) portfolio will be longer than the duration of the liability portfolio (which is likely to be made up of saving bank deposits, current account balances and short term fixed deposits).
- Therefore when short term interest rates increase, the cost of deposits will increase. The long term interest rates may remain unchanged or even fall. This can result in the interest earned on the assets remaining unchanged or even reduce the interest earnings.

Thus the interest rate risk can be moderate to severe.

Managing Interest Rate Risk:

- The interest rate risk can be mitigated by shortening the duration of the asset portfolio. This can be achieved by increasing the proportion of short term loans – for example by providing more crop loans – and reducing the proportion of medium / long term loans [e.g. loans for buying agricultural machinery]
- Alternatively if BRB has access to the interest rate swap market, BRB can enter into interest rate swaps where the bank pays fixed and receives floating rate interest. Then BRB is less exposed to a mis-match between its liabilities and assets.
- If BRB has access to the futures market, then BRB can maintain a short position in short interest rate futures contracts. Thus if interest rates increase (and the price of the short interest rate contract falls), then the bank will receive a profit to offset the higher borrowing costs.

Sources of Operations Risk

- Operational Risk is the risk of losses arising from failed processes, people or systems or from external events
- Operational Risk can be classified into two categories – business risk and event risk

Business risk includes all risks that are caused by transaction risk (e.g.; human errors) operational control risk (e.g.; fraud) and systems risk (e.g.; failure of IT System)

Event risk involves disaster risk (e.g.; natural disaster, war or market collapse) legal risk and regulatory risk.

- The sources of operational risk will include
 - Human errors
 - Frauds and theft
 - flaws in internal control & supervision
 - physical disasters
 - staff resource deficiencies
 - poor health & safety conditions
 - dependency on third party contractors or outsourcing
 - confidentiality & security breaches
 - IT, Telecommunication & other system failures

Managing Operational Risks

- Typically most operational risks can be managed using fairly simple strategies. Some of these strategies are discussed below:
- Employment - relate people risks can be managed through appropriate
 - recruitment process
 - competency management processes
 - appraisals & performance management process
 - relationship management with collective bodies like unions
- Outsourcing risks can be managed through a careful appraisal of
 - The financial standing of the third party
 - The competency, business continuity and risk-processes of the third party
- Technology related risks can be managed by
 - Keeping systems up to date
 - Routing maintenance
 - Thorough testing
 - Training staff
 - Restriction on employees' use of social media applications

(12)

[45 Marks]
